AP CALCULUS AB

Unit 1 • Lesson 1.12 practice • Answer key

Lesson 1.12 • Confirming Continuity over an Interval

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

On which interval is $f(x) = \sqrt{x-3}$ continuous?

(A) $[3, \infty)$
(B) $(3, \infty)$
(C) $(-\infty, 3]$
(D) All real numbers

Solution

Answer: (A). The square root is continuous on its domain $x \geq 3$ (one-sided at $x = 3$).

Question 2

NO CALCULATOR

Where is $f(x) = \dfrac{x+1}{x^2-4}$ continuous?

(A) All real numbers
(B) All real numbers except ± 2
(C) All real numbers except -1
(D) All real numbers except 2

Solution

Answer: (B). A rational function is continuous except where the denominator is zero.

Question 3

NO CALCULATOR

On which interval is $f(x) = \ln(x+5)$ continuous?

(A) $(0, \infty)$
(B) $[-5, \infty)$
(C) $(-5, \infty)$
(D) $(5, \infty)$

Solution

Answer: (C). ln requires $x + 5 > 0$.

Question 4

NO CALCULATOR

On which interval is $\tan x$ continuous?

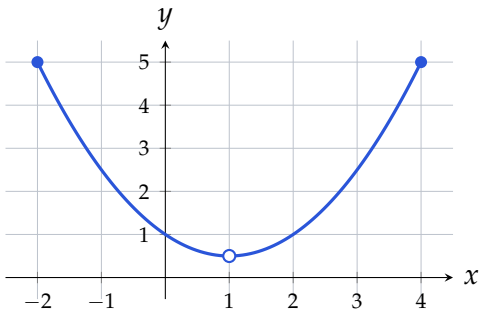
(A) $[0, \pi]$
(B) $(0, \pi)$
(C) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
(D) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Solution
Answer: (D). $\tan x$ is undefined at $\pm\frac{\pi}{2}$ and continuous between them.

Question 5 (graph)

NO CALCULATOR

The graph of f on $[-2, 4]$ is shown; it has a hole at $x = 1$. On which of the following intervals is f continuous?



(A) $[-2, 0]$
(B) $[0, 2]$
(C) $[-2, 4]$
(D) $[1, 3]$

Solution
Answer: (A). f is undefined at $x = 1$, so it is not continuous on any interval that contains 1. Only $[-2, 0]$ avoids $x = 1$.

Question 6 (Free response)

NO CALCULATOR

Let $h(x) = \dfrac{\ln x}{x^2 - 9}$.

(a) Find the domain of h, and explain why h is continuous on its domain.
(b) Is h continuous on the closed interval $[1, 2]$? On $[2, 4]$? Explain.

Solution
(a) $\ln x$ needs $x > 0$, and the denominator is zero at $x = \pm 3$. Domain: $(0, 3) \cup (3, \infty)$. h is a quotient of continuous functions with a nonzero denominator on this set, so it is continuous there.
(b) $[1, 2]$ is inside the domain, so h is continuous on $[1, 2]$. $[2, 4]$ contains $x = 3$, where h is undefined, so h is not continuous on $[2, 4]$.